

Notes on Group Theory

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Chapter 1

Introduction

Definition 1 (Symmetry). Symmetry means a bijection from some mathematical object to itself which preserves the structure of the object.

Chapter 2

Cayley's Theorem

Chapter 3

Homomorphism

3.1 Definitions

Definition 2 (Homomorphism). A homomorphism from one group G to another group H is a map $f : G \rightarrow H$ that preserves the group multiplication. This means, $\forall a, b \in G, f(a \cdot b) = f(a) \cdot f(b)$.

Remark 1. A consequence of homomorphism is that it also preserves the identity and inverse elements.

1. $f(1_G) = 1_H$
2. $f(a^{-1}) = (f(a))^{-1}$

Therefore a homomorphism can be thought of a function that preserves the group structure.

Definition 3 (Isomorphism). An isomorphism is a homomorphism which is also a bijection. Therefore, there exists an inverse map $f^{-1} : H \rightarrow G$.

Remark 2. An isomorphism between two groups means that they are in some sense the same group, except that the elements has been relabelled.

Definition 4 (Automorphism). An automorphism is a isomorphism of a group to itself, i.e. $f : G \rightarrow G$.

Remark 3. An automorphism of a group can be thought of the symmetries of the group. Recall that symmetry means a bijection from some mathematical object to itself which preserves the structure of the object. In particular, the set of all automorphisms of a group is itself a group because it is the symmetry of something.

Definition 5 (Kernel). A kernel is the set of elements on a group which maps to the identity element under a map, i.e. $\text{Ker}(G) := \{a \in G \mid f(a) = 1_H\}$.

3.2 Examples

Example 1 (Exponential map).

$$\exp(x) := 1 + x + \frac{x^2}{2!} + \cdots$$

This has the property that

$$\exp(x + y) = \exp(x) \cdot \exp(y).$$

On the left side, we have the group operation of $\mathbb{R}(+)$ and on the right, we have the group operation of $\mathbb{R}^*(\cdot)$, where $\mathbb{R}^* := \mathbb{R} \setminus \{0\}$.

It is not a bijection because it is not onto. However, it is an isomorphism from $\mathbb{R}(+)$ to $\mathbb{R}_{>0}(\cdot)$. The inverse map is $\log$.

Example 2 (Determinant of a matrix).

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

This has the property that

$$\det(AB) = \det(A) \det(B).$$

This means that the $\det$ is a homomorphism from the generalized linear group of size $n(\geq 1)$ over reals, $\text{GL}_n(\mathbb{R})$, to $\mathbb{R}^*(\cdot)$. Recall that $\text{GL}_n(\mathbb{R})$ is the set of **invertible** linear maps from $\mathbb{R}^n \rightarrow \mathbb{R}^n$.

The **kernel** in this case is $\text{SL}_n(\mathbb{R})$ (special linear group) which is defined as the set of matrices with $\det 1$.

Remark 4. In linear algebra, “special” often means that the $\det$ is 1. Therefore, the special orthogonal group of matrices means the group of orthogonal matrices which has $\det 1$. (???)

Example 3. Recall that $\mathbb{Z}/4\mathbb{Z}$ is the group of integers mod 4. The elements are $\{0, 1, 2, 3\}$ and the group operation is given by $+$. Similarly, $\mathbb{Z}/5\mathbb{Z}^*$ is the nonzero group of integers mod 5, where the elements are $\{1, 2, 3, 4\}$ and the group operation is $\cdot$. Then a map $f : \mathbb{Z}/4\mathbb{Z} \rightarrow \mathbb{Z}/5\mathbb{Z}^*$ can be defined such a way that $f(a) = 2^a$.